

A Single-Phase Single-Stage Current Source Inverter With Buck-Boost Capability and Reduced Inductor

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Abstract—For the conventional single-phase current source inverter (CSI), a large inductor is needed to stabilize the input current, which increases system volume, cost, and losses. In this article, a single-phase single-stage CSI with its operation principle is proposed. Different from conventional CSIs where only voltage-boost operation is supported, the input dc-link voltage of the proposed CSI can be higher than the peak value of output voltage (buck mode), and it can also be lower than the peak value of output voltage (boost mode). So, it is very suitable for occasions with a wide input voltage range. Moreover, the input inductor current can be balanced within per switching cycle using the proposed operation strategy. As a result, the input inductor and output filter capacitor are decreased. First, the derivation and operating modes of the CSI are given in detail. Then, the design guidelines of the components are presented, and a comparison with some existing CSIs has been made. A 1 kW prototype is also built and tested in the laboratory, the experimental results are presented to verify the effectiveness of the proposed CSI and control strategy.

Index Terms—Buck-boost inverter, current source inverter (CSI), pulswidth modulated power converter.

I. INTRODUCTION

OVER the past few years, power inverter techniques have developed rapidly due to the development of power electronics including circuit topologies, switching power devices, and control strategies [1]. Usually, inverters can be divided into voltage source inverters (VSI) and current source inverters (CSI) according to the type of input source, and all of them have been applied to industrial and civil applications.

VSI has always been treated as an attractive dc/ac inverter. It is adopted in both high-power applications [2], [3], and low-power applications [4], [5], [6]. Generally, VSIs are buck converters, and the dc-link voltage should be higher than the amplitude of the ac output voltage. Thus, a boost converter is

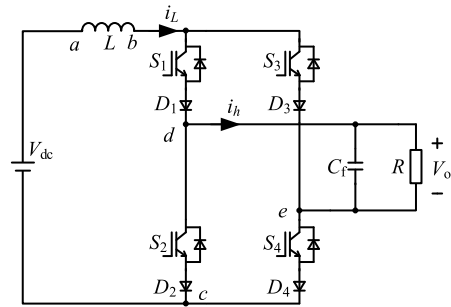


Fig. 1. Conventional single-phase CSI topology.

often required at the front end [7]. Although VSI has many advantages, some inherent drawbacks are worth noting. For example, no voltage short circuit allowed [8] large dc-link capacitor [9]. When the switching frequency is high, VSI may have electromagnetic interference (EMI) issues due to the large dv/dt [10]. Compared with an inductor, the electrolytic capacitor used in the dc-link is more prone to failure and degrades the whole system's lifetime due to the effect of ripple current, switching frequency, installation form, and heat dissipation on capacitor reliability [11], [12], [13], [14].

Compared with VSI, conventional CSI increases topology complexity due to the presence of additional diodes and dc-link inductors, as shown in Fig. 1. Additional components bring also some advantages. For example, the inductor in CSI is mechanically more robust and can be designed to withstand high temperatures [15], [16], [17], [18]. Thus, the CSI has the inherent current limiting capability, and it can enhance the system reliability. CSI performs high reliability due to the protection of short circuits [10], [19]. CSI has low EMI, and dv/dt stress over the load [20].

However, the conventional single-phase CSI has essential defects. The inductor current always increases when the dc-link voltage is higher than the absolute value of the ac output voltage. Thus, the inductance core tends to saturate as the inductor current increases. As a result, a large inductor is used as current sources in practical applications. In order to overcome the inherent defect of conventional CSI, many improved topologies and modulation methods have been proposed in the literature. A universal single-phase inverter is proposed in [21]. Though this inverter is a two-stage topology, it achieves a one-stage transformation of energy. Moreover, it can work in the working mode of buck, boost, and buck-boost. However, this inverter operates discontinuous current mode (DCM). Although it greatly reduces dc-link inductor, intermittent inductor current stress is a cause for concern.

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The topology used in [22] and [23] is similar to that in [21], but operates in a continuous inductor current mode. However, this method greatly increases the value of the inductor while achieving a wide range of input voltage. A dc capacitor is used in [24], this inverter uses the capacitor to supply the 100 Hz current ripple wave. This method can minimize the fluctuation of inductance current as much as possible within a switching cycle. However, the value of the inductor is still relatively high at a switching frequency of 10 kHz. Zhang et al. [25] proposes an inverter with an inductive current relief circuit. By combining the dc-link with the power transistor and diode, the “freewheeling model” is added to achieve stable inductance current and reduce inductance value. This topology has a large input voltage range, but it can only operate in boost mode. A CSI with improved multiloop control is presented in [26]. By introducing virtual impedance into the multiloop control strategy, the dc side current ripple is reduced, thereby reducing the inductance value of the dc side inductance. However, this topology only works on the boost condition, and the efficiency is low at rated power. Ding et al. [27] introduces the concept of VSI’s flying capacitor into CSI and achieves multilevel dc-link voltage by adding flying capacitors on the ac output side, thereby reducing dv/dt and reducing the inductance of the inductor. However, the working state of the voltage at both ends of the inductor in this topology is similar to that of traditional methods, and it can only operate in boost mode. The literature mentioned above seems difficult to meet the requirements of buck mode and the relatively small value of inductor under continuous current and low switching frequency operating conditions.

In order to overcome the drawback that the input inductor current of the conventional CSI cannot be balanced within the switching frequency and achieve both buck and boost operation, a novel single-phase CSI topology and its operation modes are proposed in this article. First, the input inductor and output filter capacitor are reduced compared with conventional CSI. Second, the proposed CSI can operate in both buck and boost configurations, so it is very suitable for wide input voltage applications. Third, the efficiency of the proposed CSI is similar to that of [23], [24], [25], [26] and conventional CSI. The rest of the article is organized as follows. Section II explains the derivation of the proposed CSI by the equivalent inductor model. Section III gives the operation modes and control strategy. The design guidelines for the input inductor and output filter capacitor, losses analysis, and comparison are shown in Section IV. Section V presents the experimental results and Section VI elaborates the conclusion.

II. DERIVATION OF THE IMPROVED CSI

A. Unified Model of Traditional CSI

For conventional single-phase full-bridge CSI, the inductor-based equivalent model is shown in Fig. 2. L is the inductor, there are three potentials V_{dc} , $|V_o(t)|$ and 0, they are connected to the terminals y , x_1 , and x_2 . Terminal a is always connected to V_{dc} , and the potentials of b are connected to points x_1 and x_2 through a single-pole double-throw switch. In order to satisfy the volt-second balance principle, the

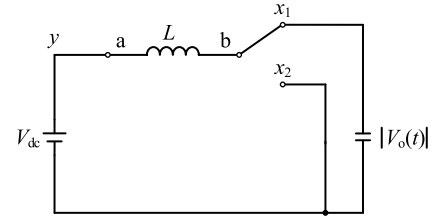


Fig. 2. Equivalent inductor model of conventional CSI.

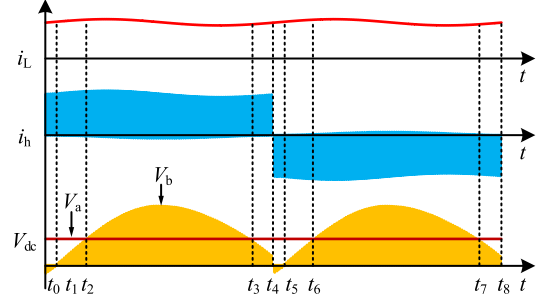


Fig. 3. Voltage and current across the dc-link inductor.

average potential of terminal a should be equal to the average potential of terminal b , thus the following equation can be obtained. Here, d is the ratio of the time when terminal b connects to point x_2 to the sum of the time when it connects to points x_1 and x_2

$$V_{dc} = d \times 0 + (1 - d)|V_o(t)|. \quad (1)$$

Assuming the point c is 0 V in Fig. 1. The voltage and current across the inductor are shown in Fig. 3. i_L is the inductor current, i_h is the bridge current, and V_a and V_b are the potentials of points a and b . During $[t_0-t_2]$ and $[t_3-t_4]$, V_a is always higher than V_b , so i_L is always increases. Similarly, i_L always increases during $[t_4-t_6]$ and $[t_7-t_8]$. Only during $[t_2-t_3]$ and $[t_6-t_7]$, V_a is between the potentials of x_1 and x_2 , and i_L is balanced within one switching period. In other words, (1) can be satisfied only during $[t_2-t_3]$ and $[t_6-t_7]$. In order to prevent magnetic saturation caused by the continuous increase of inductance current, a very large inductance has to be used on the dc-link side. It is worth noting that at times t_0-t_1 and t_4-t_5 , V_b is negative relative to point c . Due to the characteristics of the voltage phase lag current of the resistive and capacitive load and the property that the filtering capacitor C_f voltage cannot suddenly change, when the conduction sequence of the t_0 (or t_4) power switches changes, V_b is equal to V_o (or V_b is equals to $-V_o$).

Finally, the double power line frequency current fluctuation will appear in i_L , which will introduce abundant harmonics in the output current. In order to reduce the current fluctuation, the large dc-link inductor and filter capacitor are required. It undoubtedly increases the cost and volume and reduces the power density of the system.

The topology of [25], [26], and [27] can only work in boost mode and can be explained by this theory. Taking [27] as an example, an equivalent inductor model of [27] is shown in Fig. 4. According to the volt-second balance principle, the following equation can be deduced:

$$V_{dc} = d_1 \times 0 + d_2|V_o(t)| + d_3|V_o(t)/2|$$

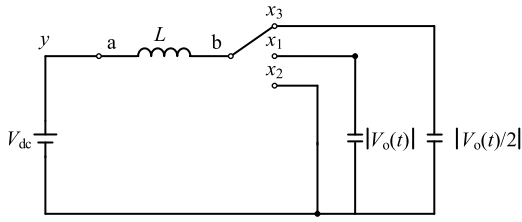


Fig. 4. Equivalent inductor model of [27].

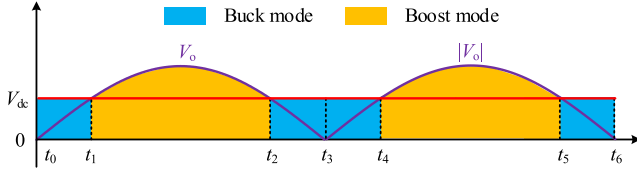


Fig. 5. Idea of bucking the input voltage.

$$= (d_2 + d_3/2)|V_o(t)| \quad (2)$$

where d_1 , d_2 , and d_3 are the ratio of the time when terminal b connects to points x_2 , x_1 , and x_3 to the sum of the time when it connects to points x_1 , x_2 , and x_3 , respectively. Obviously, the $d_2 + d_3/2$ in (2) is significantly smaller than 1, so this topology can only operate in boost mode.

B. Derivation of the Proposed CSI

Based on the above analysis, it can be concluded that the input inductor of conventional CSI is always charged when the input voltage is greater than the instantaneous value of the output voltage. In order to solve this problem, two ideas are obtained. The first is to buck the input voltage, and the second is to boost the output voltage.

The idea of bucking the input voltage is shown in Fig. 5. The red line represents the input dc-link voltage V_{dc} . The purple line represents the absolute value of the output voltage $|V_o(t)|$. During $[t_0-t_1]$, $[t_2-t_4]$, and $[t_5-t_6]$, the dc-link voltage is bucked to a level less than or equal to the output voltage. The equivalent inductor model is shown in Fig. 6. An additional terminal y_2 is introduced in the input stage, and the potential of y_2 is 0. According to the volt-second balance principle, the following equation can be obtained:

$$V_{dc}d_1 = |V_o(t)|d_2 \quad (3)$$

where d_1 is the ratio of the time when terminal a connects to point y_1 to the sum of the time when it connects to points y_1 and y_2 , d_2 is the ratio of the time when terminal b connects to point x_1 to the sum of the time when it connects to points x_1 and x_2 . In the blue area, where the inverter operates in buck mode, the voltage level of terminal a continuously changes between V_{dc} and 0 through different duty cycles, while adjusting d_2 at the same time. This can make the average potential of the terminals on both ends of the inductor equal in one output cycle, and the inductor current can be balanced in the switching frequency. Compared to traditional CSI, this method can reduce the parameters of inductor L . Based on Fig. 6, the new topology can be realized by adding a buck converter to the front end of conventional CSI. However,

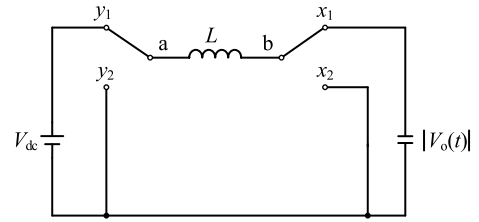


Fig. 6. Equivalent inductor model of bucking the input voltage.

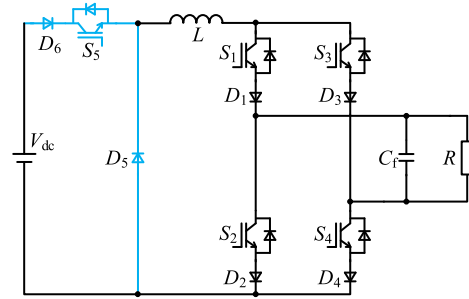


Fig. 7. CSI topology with bucking the input voltage.

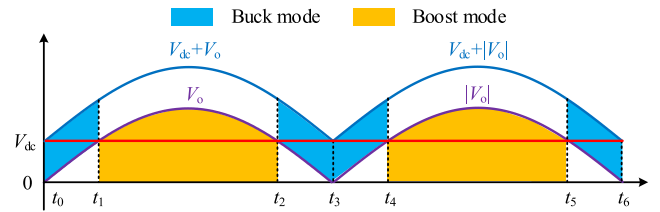


Fig. 8. Idea of boosting the output voltage.

it is still a two-stage topology. On the basis of the two-stage inverter, a new single-stage CSI is proposed by appropriately altering the PWM control as shown in Fig. 7. The performance of the inverter in DCM and CCM were verified in [21], [22], and [23], respectively.

The idea of boosting the output voltage is shown in Fig. 8. During $[t_0-t_1]$, $[t_2-t_4]$, and $[t_5-t_6]$, add an output level with an absolute value of $V_{dc} + |V_o(t)|$, as shown by the blue line in Fig. 8. Based on Fig. 8, the equivalent inductor model is derived as shown in Fig. 9. An additional terminals x_3 is introduced, and the potentials for it is $V_{dc} + |V_o(t)|$. During $[t_0-t_1]$ and $[t_2-t_3]$, terminal b switches connections between x_1 and x_3 . As represented as a blue area in Fig. 8, during $[t_1-t_2]$, terminal b switches connections between x_1 and x_2 and is represented as a yellow area in Fig. 8. By doing this, the potential of terminal b can always be maintained at two levels, one higher than itself and the other lower than itself, and the volt-second principle of L is always satisfied. As a result, L can be reduced. In addition, the CSI operates under buck configuration during $[t_0-t_1]$ and $[t_2-t_3]$ because V_{dc} is higher than $V_o(t)$, and it operates under boost configuration during $[t_1-t_2]$ because V_{dc} is lower than $V_o(t)$. Therefore, the CSI can not only reduce the inductor but also has the buck-boost ability, and it is very suitable for wide input voltage applications.

Based on Fig. 9, the circuit schematic of the new CSI is shown in Fig. 10. Six switches (S_1-S_6) and six diodes (D_1-

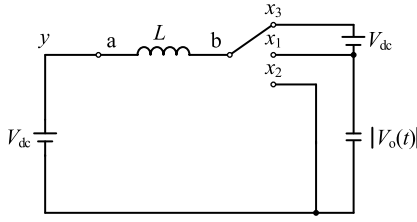


Fig. 9. Equivalent inductor model of boosting the output voltage.

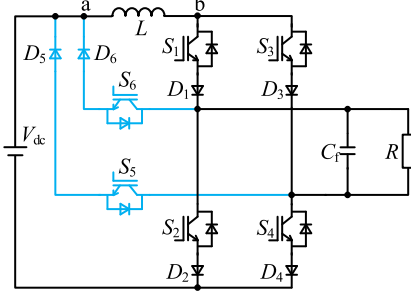


Fig. 10. Proposed single-phase CSI topology.

D_6) are required. C_f is the output filter capacitor and R is the load. Compared with conventional single-phase CSI, two groups of branches (S_5, D_5) and (S_6, D_6) are added to the circuit. These two branches are connected to the positive bus and different midpoints of bridge arms. Although additional switches and diodes are added, the current always flows two switches and two diodes as conventional CSI, it belongs to a single-stage topology.

III. OPERATION MODES AND CONTROL STRATEGY

A. Operation Modes of the Proposed CSI

There are six switching modes for the proposed CSI, and Fig. 11 shows the state of all switches in each mode. Here, the red line represents the current paths. Assuming point c is at zero potential.

Mode A: Switches S_1 and S_2 are turned on when $V_o(t)$ is positive, as shown in Fig. 11(a). Terminal a is connected to the positive of the dc-link and V_a is V_{dc} . Terminal b is connected to the negative of the dc-link and V_b is zero. In this case, L is charged by the dc-link voltage and i_L is increased.

Mode B: Switches S_1 and S_5 are turned on when $V_o(t)$ is positive and the dc-link voltage is higher than the output voltage ($V_{dc} > V_o(t)$), as shown in Fig. 11(b). Terminal a is connected to the positive of the dc-link and V_a is V_{dc} . Terminal b is connected to the positive of C_f . The negative of C_f is connected to the positive of the dc-link, so V_b is $V_{dc} + V_o(t)$. In this case, the inductor is discharged by the negative inductor voltage V_{ab} and i_L is decreased.

Mode C: Switches S_1 and S_4 are turned on when $V_o(t)$ is positive as shown in Fig. 11(c). Terminal a is connected to the positive of the dc-link and V_a is V_{dc} . Terminal b is connected to the positive of C_f . The negative of C_f is connected to the negative of the dc-link and V_b is $V_o(t)$. When $V_o(t) > V_{dc}$, the inductance voltage $V_{ba} > 0$. At this time, the inductor voltage and current are in the same direction, and the inductor current

TABLE I
SUMMARY OF THE CHARGING AND DISCHARGING
STATE OF THE INDUCTOR

Modes	V_a	V_b	V_{ab}	i_L
A	V_{dc}	0	V_{dc}	$\uparrow$
B	V_{dc}	$V_{dc} + V_o(t)$	$-V_o(t)$	$\downarrow$
C	V_{dc}	$V_o(t)$	$V_{dc} - V_o(t)$	$\uparrow$ or $\downarrow$
D	V_{dc}	0	V_{dc}	$\uparrow$
E	V_{dc}	$V_{dc} - V_o(t)$	$V_o(t)$	$\downarrow$
F	V_{dc}	$-V_o(t)$	$V_{dc} + V_o(t)$	$\uparrow$ or $\downarrow$

$\uparrow$ represents charging and $\downarrow$ represents discharging.

decreases. When $V_o(t) < V_{dc}$, the inductor voltage $V_{ba} < 0$. At this time, the inductor voltage and current are reversed, and the inductor current increases.

Mode D: Switches S_3 and S_4 are turned on when $V_o(t)$ is negative, as shown in Fig. 11(d). Terminal b is connected to the negative of the dc-link and V_b is zero. Terminal a is connected to the positive of the dc-link and V_a is V_{dc} . In this case, the inductor is charged by the positive voltage V_{dc} and i_L is increased.

Mode E: Switches S_3 and S_6 are turned on when $V_o(t)$ is negative and dc-link voltage is higher than $|V_o(t)|$ ($V_{dc} > |V_o(t)|$) as shown in Fig. 11(e). Terminal a is connected to the positive of the dc-link and V_a is V_{dc} . Terminal b is connected to the negative of C_f . Since the positive of C_f is connected to the positive of the dc-link, V_b is $V_{dc} - V_o(t)$. In this case, the inductor is discharged by $V_o(t)$, i_L is decreased.

Mode F: Switches S_2 and S_3 are turned on when $V_o(t)$ is negative as shown in Fig. 11(f). Terminal b is connected to the negative of C_f . The positive of C_f is connected to the negative of the dc-link and V_b is $-V_o(t)$. Terminal a is connected to the positive of dc-link and V_a is V_{dc} . When $|V_o(t)| > V_{dc}$, the inductance voltage $V_{ba} > 0$. At this time, the inductor voltage and current are in the same direction, and the inductor current decreases. When $|V_o(t)| < V_{dc}$, the inductor voltage $V_{ba} < 0$. At this time, the inductor voltage and current are reversed, and the inductor current increases.

B. Control Strategy and Gate Signals Generation

Table I summarizes the voltage across the inductor and its influence on the inductor current in all modes. As we can see, i_L is increased in Modes A and D, and it is decreased in Modes B and E. In Modes C and F, the increase or decrease of i_L depends on the relative magnitude of V_{dc} and $V_o(t)$. By the reasonable selection of the modes, the inductor current can be balanced within the switching cycle. For example, during $[t_0-t_1]$ and $[t_2-t_3]$ in Fig. 8, Modes B and C can be selected to balance the inductor current; during $[t_1-t_2]$, Modes A and C can be selected to balance the inductor current. In other intervals, the principle of mode selection is similar.

The proposed CSI is derived from the conventional CSI; thus, the mathematical modeling of the proposed CSI is very similar to those of conventional CSI. The voltage controller used in conventional CSI is applied to the proposed CSI. Similar to conventional CSI, the gate signal of the proposed CSI also requires the addition of a dead time to ensure a complete freewheeling circuit for the inductor current. Fig. 12

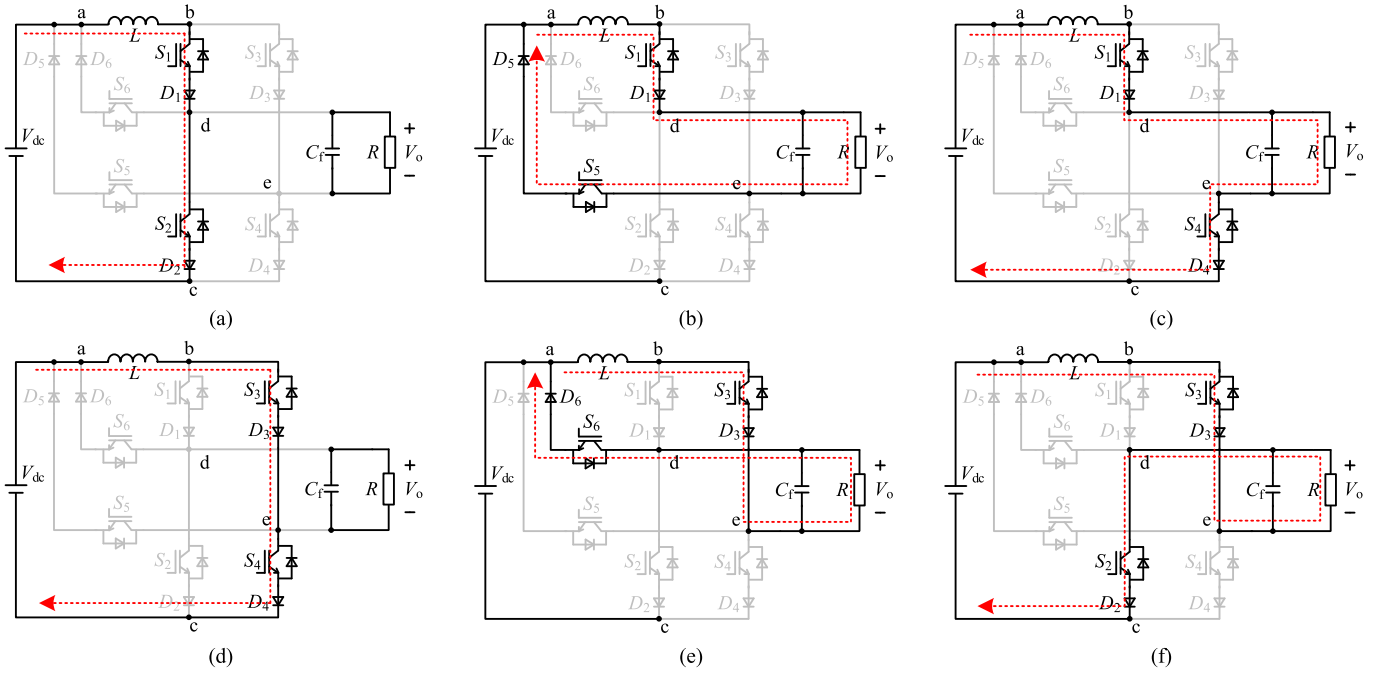


Fig. 11. Six modes of the proposed CSI topology. (a) Mode A: $V_a = V_{dc}$, $V_b = 0$. (b) Mode B: $V_a = V_{dc}$, $V_b = V_{dc} + V_o(t)$. (c) Mode C: $V_a = V_{dc}$, $V_b = V_o(t)$. (d) Mode D: $V_a = V_{dc}$, $V_b = 0$. (e) Mode E: $V_a = V_{dc}$, $V_b = V_{dc} - V_o(t)$. (f) Mode F: $V_a = V_{dc}$, $V_b = -V_o(t)$.

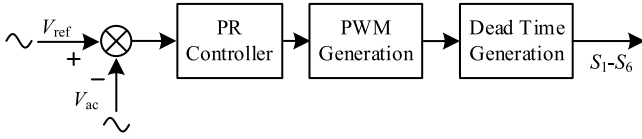


Fig. 12. Control block diagram for the proposed CSI.

shows the control block diagram for the proposed CSI. V_{ref} is the reference sine wave. Given that the proportional resonant (PR) controller has better tracking performance for ac signals than a PI controller, the PR controller is chosen to adjust the output ac voltage [28]. The PWM generation module in Fig. 12 is used to generate the gate signals for six switches. Here, the pulse widths are modulated by comparing a reference waveform with a high-frequency triangular waveform. The references for six switches are shown in Fig. 13, and the corresponding duty cycle of each switch is shown in Table II according to the volt-second balance principle. Where M is the ratio of V_o to V_{dc} , θ is the angle of V_o , and d is the duty cycle of the switching tube with respect to time variation. When a switching cycle is small enough, the curve in Fig. 13 becomes smoother rather than a step change. The dead time generation module is used to generate signals that require at least two power switches to conduct at any time.

C. Voltage and Current Analysis

Take Mode A as an example. In this mode, $V_a = V_{dc}$, $V_b = V_c = V_d = 0$. $V_e = -V_o(t)$, the voltage stress of switch S_1, S_2 and diode D_1 , and D_2 is 0. For the branch of S_3 and D_3 , to satisfy the switch S_3 and diode D_3 are turned off, the midpoint potential of both should be the same as point e . So, the voltage stress of switch S_3 is $V_o(t)$ and diode

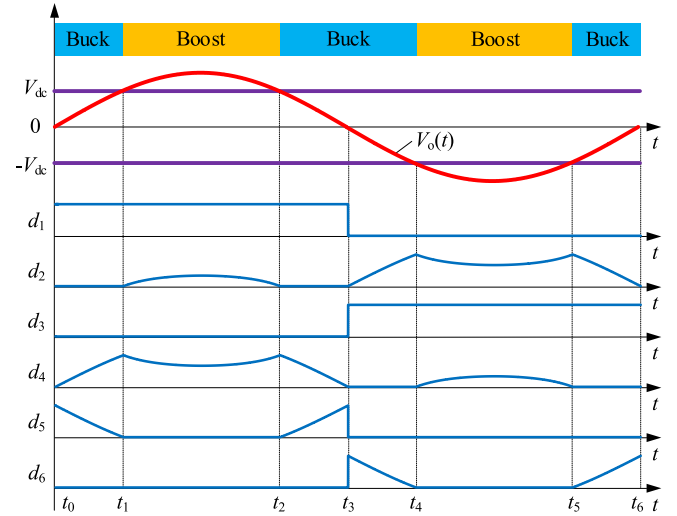


Fig. 13. References for six switches during a power-frequency cycle.

TABLE II
DUTY RATION OF EACH SWITCH

	$t_0 \sim t_1$	$t_1 \sim t_2$	$t_2 \sim t_3$	$t_3 \sim t_4$	$t_4 \sim t_5$	$t_5 \sim t_6$
S_1	1	1	1	0	0	0
S_2	0	$1-1/M\sin\theta$	0	$M \sin\theta $	$1/M \sin\theta $	$M \sin\theta $
S_3	0	0	0	1	1	1
S_4	$M\sin\theta$	$1/M\sin\theta$	$M\sin\theta$	0	$1-1/M \sin\theta $	0
S_5	$1-M\sin\theta$	0	$1-M\sin\theta$	0	0	0
S_6	0	0	0	$1-M \sin\theta $	0	$1-M \sin\theta $

D_3 is 0. By using the same principles, it can be obtained that the voltage stress of power switches of S_4, S_5 , and S_6 is 0, and the voltage stress of diodes of D_4, D_5 , and D_6 is V_o , $V_{dc} + V_o(t)$, and V_{in} . Through analyzing all six modes, the

TABLE III
MAXIMUM OF VOLTAGE AND CORRESPONDING MODES

Voltage stress	The mode in which the maximum voltage stress occurs
V_{S1}	$ V_o(t) $ Mode D
V_{D1}	0 Mode A
V_{S2}	$V_{dc} + V_o(t)$ Mode B
V_{D2}	$ V_o(t) $ Mode D
V_{S3}	$V_o(t)$ Mode A
V_{D3}	0 Mode A
V_{S4}	$V_{dc} + V_o(t) $ Mode E
V_{D4}	V_o Mode A
V_{S5}	$ V_o(t) $ Mode E
V_{D5}	$V_{dc} + V_o(t)$ Mode A
V_{S6}	$V_o(t)$ Mode B
V_{D6}	$V_{dc} + V_o(t) $ Mode D

maximum voltage stress of power switches and diodes and their corresponding modes are shown in Table III.

Because of the additional new modes (Mode B, Mode E), the voltage stress of S_2 and S_4 increases to $V_{dc} + V_o(t)$ from $V_o(t)$. At the same time, the maximum voltage of diode D_5 and D_6 occurs in Mode A and Mode D, while the maximum voltage of two new power switches S_5 and S_6 is the same as S_1 and S_3 .

When a power switch or diode is turned on, the current stress is the inductor current i_L , with the difference in the constant value of the inductor current of the conventional CSI, the inductor current of the proposed CSI fluctuates. Assuming the input power and the input voltage are the same, the maximum value of the inductor current of the proposed CSI is higher than conventional CSI.

IV. DESIGN GUIDELINES AND COMPREHENSIVE ANALYSIS

A. DC-Link Inductor Design Guideline

The inductor design is decided by its maximum current ripple. Usually, the current ripple (ΔI) of an inductor is related to the voltage across it (ΔU) and the operation time (Δt) as expressed with the following equation:

$$\Delta I = \frac{\Delta U}{L} \Delta t. \quad (4)$$

When V_{dc} is lower than V_m , the CSI needs to shuffle between buck and boost modes frequently. Combining (4) and switching modes in Fig. 11, the input inductor current ripple $\Delta i_L(t)$ in both buck and boost modes can be expressed as the following equations, respectively. Here, f_s is the switching frequency

$$\Delta i_L(t) = \frac{(V_{dc} - |V_o(t)|)|V_o(t)|}{Lf_s V_{dc}} \quad (5)$$

$$\Delta i_L(t) = \frac{V_{dc}(|V_o(t)| - V_{dc})}{Lf_s |V_o(t)|}. \quad (6)$$

Based on (5) and (6), the maximum value of inductor current ripple $\Delta I_{L \max}$ in both buck and boost modes are shown in the following equations, respectively,

$$\Delta I_{L \max} = \frac{V_{dc}}{4Lf_s} \quad (7)$$

$$\Delta I_{L \max} = \frac{V_{dc}(V_m - V_{dc})}{Lf_s V_m}. \quad (8)$$

Clearly, in buck mode, the maximum current ripple occurs when $|V_o(t)| = V_{dc}/2$. The minimum current ripple is 0, and it occurs when $|V_o(t)| = 0$ or V_{dc} . In boost mode, the maximum current ripple occurs when $|V_o(t)| = V_m$. The minimum current ripple occurs when $|V_o(t)| = V_{dc}$ and the value of the minimum current ripple is also zero. To meet the requirement of the current ripple, the input inductor should satisfy the following equation:

$$L \geq \frac{V_{dc}}{f_s \Delta I_{L \max}} \max \left\{ \frac{1}{4}, 1 - \frac{V_{dc}}{V_m} \right\}. \quad (9)$$

At the same time, the inverter needs to operate in the continuous conduction mode (CCM). Then the first equation that follows can be obtained, where $I_L(t)$ is the average value of inductor current in each switching period, it can be obtained by the instantaneous power in the second equation that follows:

$$\frac{1}{2} \Delta i_L(t) \leq I_L(t) \quad (10)$$

$$I_L(t) = \frac{|V_o(t)||i_o(t)|}{V_{dc}}. \quad (11)$$

Then, according to (5), (6), (10), and (11), the input inductor should satisfy the following equation:

$$L \geq \frac{1}{2f_s} \max \left\{ \frac{(V_{dc} - |V_o(t)|)|V_o(t)|}{|i_o(t)|}, \frac{V_{dc}^2(|V_o(t)| - V_{dc})}{(V_o(t))^2|i_o(t)|} \right\}. \quad (12)$$

Finally, the input inductor needs to satisfy both (9) and (12).

B. Output Capacitor Design

The output capacitor C_f is designed to eliminate harmonic current caused by the actions of the switches. In the circuit working in buck mode, which is in Fig. 11(b) and (c) or (e) and (f), the inductor L and output filtering capacitor C_f form an LC filter. At this point, the parameters of C_f depend on the cut-off frequency. Then, the output capacitor C_f can be roughly expressed as the following equation:

$$f_c = \frac{1}{2\pi\sqrt{LC_f}}. \quad (13)$$

When the circuit operates in boost mode, which is shown in Fig. 11(a) and (c) or (d) and (f), the parameters of the output capacitor should take into account the maximum ripple of the output voltage. It can be expressed as the following equations:

$$P_o \Delta t = C_f [(V_m + \Delta V)^2 - (V_m - \Delta V)^2] / 2 \quad (14)$$

$$C_f = \frac{P_o \Delta t}{2V_m \Delta V} \quad (15)$$

where P_o is the working power, Δt is the discharge time, V_m is the output voltage, and ΔV is the output voltage ripple.

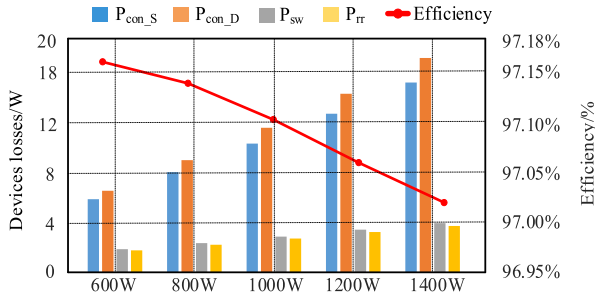


Fig. 14. 200 V fixed input voltage loss distribution and efficiency curve.

C. Losses Analysis

During the conduction mode, the IGBT collector-emitter voltage drop v_{ce} is approximated as the following equation [29]:

$$v_{ce}(t) = v_{ce0} + R_c i_c(t) \quad (16)$$

where v_{ce0} is the ON-state collector-emitter forward voltage drop and R_c is the collector-emitter ON-state resistance.

For antiparallel diodes, the voltage drop is calculated as

$$v_F(t) = v_{F0} + R_F i_F(t) \quad (17)$$

where v_{F0} is the ON-state forward voltage drop and R_F is the ON-state resistance

$$P_{con_IGBT}(t) = \frac{1}{2\pi} \int_0^{2\pi} [v_{ce}(t) i_c(t) d_{sw}(t)] d(\omega t) \quad (18)$$

$$P_{con_Diode}(t) = \frac{1}{2\pi} \int_0^{2\pi} [v_F(t) i_F(t) d_F(t)] d(\omega t). \quad (19)$$

The average conduction losses of IGBT (P_{con_IGBT}) and diode (P_{con_Diode}) are given as (18) and (19), respectively. i_c and i_F are the current flowing through the switch and diode and d_{sw} and d_F are the duty cycle of the switch and diode, respectively.

As has been discussed in [30], the switching losses (P_{sw}) of the switch can be obtained as

$$P_{sw} = \frac{V_{OFF} f_s (t_{on} + t_{off})}{4\pi} \int_0^{2\pi} |i(\theta)| d\theta \quad (20)$$

where V_{OFF} is the OFF-state voltage, t_{on} and t_{off} are turn-on and -off time, respectively, and i is the current flowing across switches.

The reverse recovery loss (P_{rr}) of the diode can be expressed as the following equation, Q_{rr} is the reverse recovery charge, V_R is the voltage across the diode, and I_f is the test current [30]

$$P_{rr} = \frac{1}{2\pi} \int_0^\pi \frac{i_F(t)}{I_f} Q_{rr} V_R dt \cdot f_s. \quad (21)$$

In order to show the efficiency performance, the loss and efficiency of the proposed topology are calculated. The rms value of the output voltage is 220 V, and assuming the input voltage is fixed at 200 V, the output power varies from 600 to 1400 W, and the input voltage ranges from 200 to 400 V when the output power is fixed at 1000 W. The results are shown in Figs. 14 and 15, respectively, for the

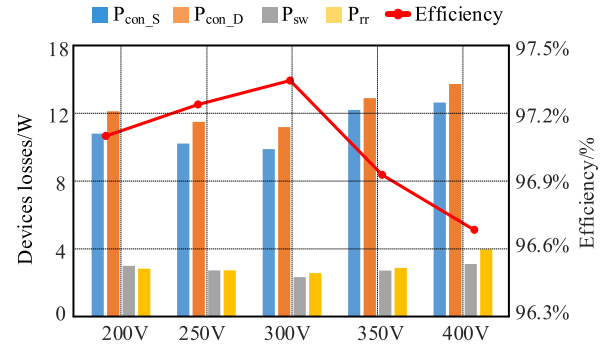


Fig. 15. 1000 W fixed output power loss distribution and efficiency curve.

TABLE IV
SYSTEM PARAMETERS

Description	Symbols	Values
Input Voltage	V_{dc}	200-400 V
Output Voltage	V_o	220 V @ 50 Hz
Input Inductance	L	3 mH
Filter Capacitor	C_f	5 μ F
Load impedance	R	50 Ω
Switching Frequency	f_s	10 kHz
IGBT	S_2, S_4	IKW15N120
	S_1, S_3, S_5, S_6	IKW20N60
Diode	D_5, D_6	IDP30E120
	D_1, D_2, D_3, D_4	IDP30E60
Power	P	968W

TABLE V
COMPARISON BETWEEN DIFFERENT TOPOLOGIES

	a	b	c	d	e	f	g
Conventional CSI	4	4	1	1	2	2	No
[24]	5	5	2	1	2/3	2/3	No
[25]	5	5	1	1	1/2	1/2	No
[26]	4	4	3	3	2/3	2/3	No
[27]	8	8	2	3	4	4	No
[21]	5	6	2	1	2/3	2/3	Yes
[23]	5	5	2	2	2/3	2/3	Yes
Proposed CSI	6	6	1	1	2	2	Yes

a. Total number of switches. b. Total number of diodes. c. Total number of inductors. d. Total number of capacitors. e. Number of conducted switches in each switching state. f. Number of conducted diodes in each switching state. g. Buck and boost capability.

proposed CSI, the selected devices, and the other parameters are listed in Table IV.

Fig. 14 shows the loss distribution of the proposed CSI at output 600 to 1400 W. It can be seen that the conduction losses of switches and diodes are the main losses. As the output power increases, under the condition of constant output voltage, the current flowing through the switches and diodes increases, thereby doubling the conduction loss. Fig. 15 shows the variation of loss distribution with input dc voltage at rated 1000 W output. In the case of both buck and boost at 300 V and below, the loss decreases with the increase of voltage. When the input voltage is greater than the maximum value of the output voltage, in the other word, that is only in buck mode, the working state of switches S_5 and S_6 is longer per unit time compared to the former, and the conduction

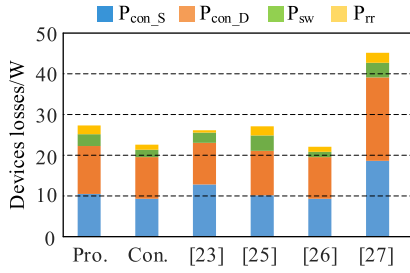


Fig. 16. Device losses in different topologies.

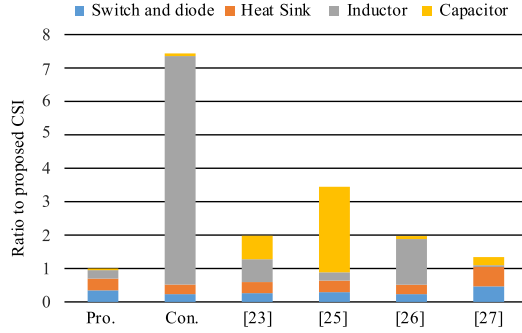


Fig. 17. Comparison of converter volumes.

losses will increase significantly. Therefore, in this operating mode, the losses will increase with the increase of voltage.

D. Comparison With Other Topologies

A comparison among several topologies has been provided in Table V. All topologies can be roughly divided into two categories according to whether they have the ability to buck and boost. Conventional CSI requires a large inductance due to its lack of self-balancing ability, such as 300 mH at 4 kHz [28]. For [24], [25], [26], [27], [28], [29], these topologies reduce the size of the dc inductance by reducing the dv/dt on both sides, such as 5 mH, 120 μ F at 2 kHz [27], 4 mH, 265 μ F at 10 kHz [25]. These topologies introduce additional switch and diode devices to meet the self-balancing of inductors or capacitors. For the topologies in [21], [22], and [23], these topologies can work in buck mode. For [21], the topology has been operating in the DCM state. Although the inductance value is relatively small (0.61 mH, 4 μ F at 10 kHz), considering the large current pulses caused by the intermittent inductance current, this to some extent offsets the advantages brought by the small inductance value.

Fig. 16 shows the loss distribution of the proposed CSI, conventional CSI, and [23], [25], [26], and [27] topology at 1 kW, 200 V input, maximum 311 V output voltage, and 10 kHz carrier frequency. It can be seen from the figure that among the total losses, conduction loss is the main loss. Compared with the conventional CSI, [23] and [26], the total loss of the proposed CSI is relatively high and similar to the topology proposed by [25]. The losses are relatively high because there are always four power switches and the diodes conduct in the topology proposed by [27].

Fig. 17 shows the volume comparison of several topologies. Assuming that the packaging and driving of power switches and diodes are the same, their volume is proportional to their

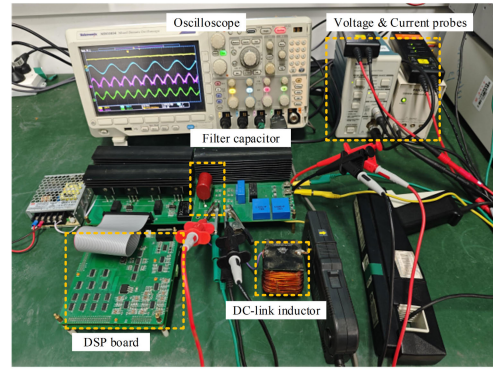


Fig. 18. Prototype of the proposed CSI.

quantity. Gurpinar and Castellazzi [31] provides a detailed calculation process for the volume of the heat sink, here assuming that the volume of the heat sink is proportional to the power loss of the device. The power losses of different topologies under the same conditions of 1 kW input power, 10 kHz carrier frequency, 200 V input voltage, and 220 V output effective voltage were calculated here. Under the condition of ensuring the same power density, the volume of inductance and capacitance is positively correlated with their maximum stored energy. The volume of the proposed CSI is unit 1, and the volume of other topologies is relative to the proposed CSI. The total volume is represented by the weighted average, in which the weighted factors of switch and diode, heat sink, inductance, and capacitance are 0.35, 0.35, 0.25, and 0.05, respectively. It can be seen that the proposed CSI has the smallest volume compared to other topologies, followed by the topology in [27]. Conventional CSI has the largest volume due to the presence of large inductors.

V. SIMULATION AND EXPERIMENTAL VERIFICATION

To validate the effectiveness of the proposed CSI, a 968 W prototype was constructed in the laboratory. The photograph of the prototype is shown in Fig. 18 and the experimental parameters are shown in Table IV. In the experimental test, a 3 mH and 15 A dc inductor is selected. The gate signals are generated in a DSP (TMS320F28335) controller. In order to verify the buck and boost capability of the CSI, some experimental tests are carried out under various dc-link voltages but the same output ac voltage.

In order to verify the performance of the proposed CSI under boost configuration, the experimental tests under different dc-link voltages are carried out. Here, the input dc-link voltage varies from 200 to 400 V. And the peak value of the output voltage is always set at 311 V. The experimental results are shown in Fig. 19. In Fig. 19(a), the input voltage V_{dc} , the output voltage V_o , the input dc-link current i_{dc} , and the input inductor current i_L are shown from top to bottom. It is observed that the dc-link voltage is 200 V, and the peak value of V_o is 311 V, thus the voltage gain is about 1.55. The input inductor current i_L is stable due to the charge and discharge balance during each switching period. As a result, the input inductor can be reduced. The maximum value of inductor current ripple occurs at the moment of the peak value of output voltage.

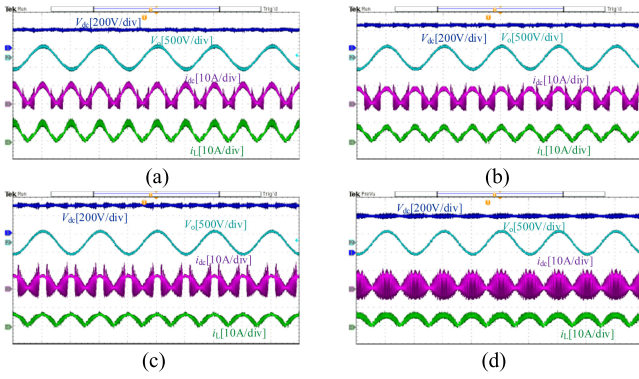


Fig. 19. Experimental waveforms under different input dc-link voltages. (a) $V_{dc} = 200$ V. (b) $V_{dc} = 250$ V. (c) $V_{dc} = 300$ V. (d) $V_{dc} = 400$ V.

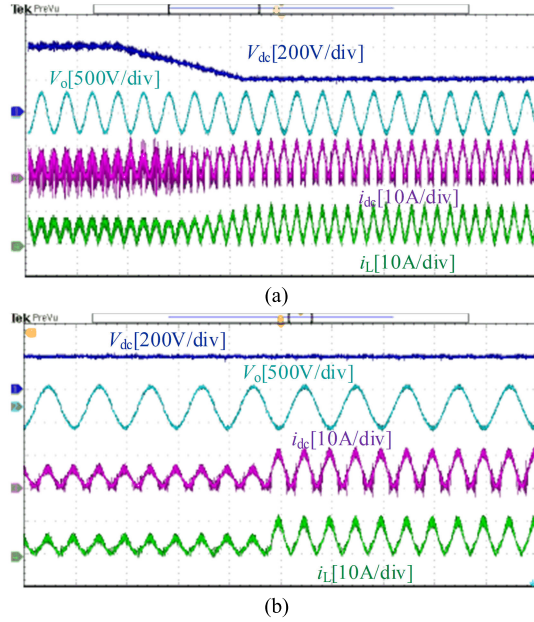


Fig. 20. Dynamic response of the CSI. (a) Step change in the value of input voltage. (b) Step change in load.

Fig. 19(b) and (c) show the experimental results when the dc-link voltages are 250 and 300 V, respectively. Obviously, as V_{dc} increases, both the rms value and current ripple of i_L decrease under the same output power rating. Therefore, the conduction losses will reduce and the overall efficiency of the CSI will increase. In order to verify the performance of the proposed CSI under buck configuration, the experimental test under a 400 V dc-link voltage is carried out. The results are shown in Fig. 19(d). It can be also observed that the peak value of the output voltage is well maintained at 311 V, thus the voltage gain is about 0.78 in this situation.

Transient responses to the proposed CSI are also conducted. In Fig. 20(a), the dc-link voltage dropped from 400 to 200 V, the CSI is switched from buck configuration to boost configuration. It can be observed that the output voltage has been stable at 220 V without any fluctuation. In Fig. 20(b), the dc-link voltage is maintained at 200 V, while the load resistor is switched from 100 to 50 Ω . It is observed that both the input and output currents are doubled, while the output voltage is well maintained at 220 V. Therefore, the proposed CSI has an excellent dynamic response, and it is very suitable for wide input voltage applications. Different from conventional CSI,

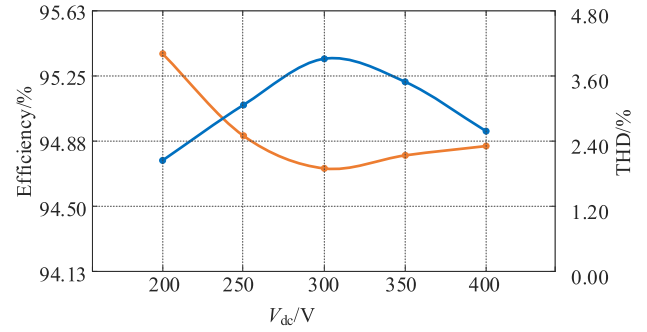


Fig. 21. Measured efficiency and THD curves with dc-link voltage.

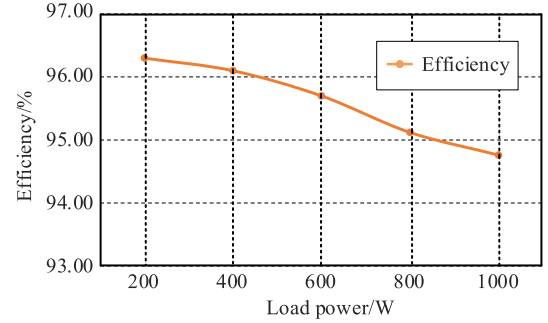


Fig. 22. Measured efficiency curve with load power.

the input current is composed of double power frequency component (100 or 120 Hz) and switching frequency ripple component, rather than a constant value. So, when the load power factor is low, the inductance current has more harmonic content, which leads to the output voltage THD not meeting the regulations.

By the power analyzer (HIOKI 3390), the measured efficiency as a function of input dc-link voltage for the proposed CSI is shown in Fig. 21, and the out power is 968 W. Seen from Fig. 21, the efficiency is gradually increased as the input voltage increases when the input voltage is less than 300 V, and the efficiency is gradually decreased as the input voltage increases when the input voltage is greater than 300 V. The measured THD of the output voltage is also shown in Fig. 21. When the input voltage deviates from the peak value of the output voltage, there will be a large ripple on the input inductor, resulting in high THD. So, the THD of output voltage achieves its minimum value when the input voltage is equal to the peak value of output voltage, which is about 311 V. The efficiency curve with load power as shown in Fig. 22. The load power is 200, 400, 600, 800, and 1000 W, respectively. The rated load power is 1000 W. When the load power increases, the efficiency of the proposed CSI decreases. When the load power reaches the rated power, the proposed CSI has the lowest efficiency, which is 94.76%. When the load power is low, the current is low, so the active devices of the proposed CSI have relatively low power losses. As the load power increases, the proportion of losses in power switches and diodes increases.

VI. CONCLUSION

A novel single-phase buck-boost CSI and its operation strategy are proposed in this article. In order to operate

under both buck and boost configurations, additional switching states are introduced. Through a reasonable combination of switching states, it can meet the voltage reduction operation where the output is always smaller than the input, so the front-end converters are no longer needed. In addition, the proposed CSI can ensure that the rise and fall of the inductor current are equal in a switching cycle. It can reduce the dc-link inductor more effectively in the continuous current mode compared to conventional buck CSI. The experimental tests show that the proposed CSI can switch smoothly between different configurations and has excellent dynamic performance. The proposed CSI can be applied in single-phase grid connection, induction heating, avionics systems, and other applications.

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